

Cloud Security - PBL

Project Execution Phases

1: Responsibility Mapping & Asset Definition (Unit 1)

Students begin by designing the architecture of a **3-tier web** application (Web, App, DB). They must create a Shared Responsibility Matrix specifically tailored to their application, documenting exactly which security controls they are responsible for versus the cloud provider.

OR

Objective

Understand how quickly a normal Internet-facing server can receive automated scans and login attempts.

Task

1. Launch an **AWS EC2 Linux instance**.
2. Configure the Security Group:

```
SSH (22) → 0.0.0.0/0  
HTTP (80) → 0.0.0.0/0
```

1. Keep the instance running for **24–48 hours**.
2. Do **not share the public IP** with anyone.
3. After 24–48 hours, check SSH logs:

```
sudo journalctl -u sshd
```

Find failed attempts:

```
sudo journalctl -u sshd | grep -i "failed"
```

Investigate

Record:

- How many failed attempts occurred?
- Which source IPs tried to connect?

- Which usernames were attempted (`root`, `admin`, etc.)?
- How long did it take to receive the first attempt?
- Were the attempts likely automated?

Final Question

Do attackers really target only big companies?

Explain your conclusion using evidence from your EC2 logs.

2: IAM & Least Privilege Enforcement (Unit 2)

Students will create a rigorous IAM structure. They must create distinct IAM Groups (Admins, Developers, Auditors) with attached custom policies following the Principle of Least Privilege. MFA must be enforced for the Admin group. A custom IAM Role must be created to allow an EC2 instance to access S3 securely without hardcoded credentials.